

NICHOLAS PARADIZOV

Mechanical Engineering · Robotics · Product Design

[City, State] · [Email] · [Phone] · [Portfolio URL] · [LinkedIn/GitHub if useful]

EDUCATION

[High School Name] — [City, State]

Expected Graduation: 2027

Relevant Coursework: AP Physics C, Calculus, AP Computer Science, Engineering

ENGINEERING & ROBOTICS EXPERIENCE

FIRST Tech Challenge Team 23268 Ultraviolet — Mechanical Design, CAD, Fabrication &

Programming

2024–Present

- Compete on a three-person FTC robotics team, contributing across CAD, prototyping, fabrication, assembly, testing, programming, and competition development.
- Designed the team's **Modular Chassis MK1**, a four-module removable mecanum drivetrain with integrated wire management and serviceable drivetrain assemblies; received a personal “**You Inspire Me**” judge pin at the FIRST World Championship for the design.
- Developing **Modular Chassis MK2**, integrating removable drivetrain cartridges, sensing, diagnostic hardware, custom PCB design, and service-oriented mechanical architecture.
- Developed custom computer-vision systems using **OpenCV and TensorFlow**, including a training pipeline built around approximately **200,000 labeled images**.
- Designed the final **V3 turret/shooter assembly** for the team's DECODE robot, adding counter-rollers for improved projectile control while reducing overall subsystem size.
- Helped manufacture custom robot components through **FDM printing, laser-cut polycarbonate, and waterjet-cut aluminum**.
- Team accomplishments include a **New Jersey State Championship, Think Award**, World Championship qualification, and worldwide competitive rankings including **4th worldwide** during Into the Deep.

SELECTED ENGINEERING PROJECTS

IDW Ratchet MK1 & MK2 — Original Transforming Figures

- Designed a complete ~40-step, 100+ part transforming figure from comic artwork with no existing toy or transformation to reverse engineer.
- Currently redesigning the figure for substantially greater source accuracy using advanced spatial packaging, surface modeling, and custom transformation mechanisms.

Custom AI Vision System — FTC Robotics

- Recorded controlled datasets, automated image labeling through adapted open-source software, and iteratively trained a TensorFlow detection model.
- Used failed testing and hardware limitations to study robustness, real-world vision conditions, and assumptions in system validation.

Mechanical Puzzle Box Series

- Designed personalized multi-step puzzle products integrating mechanical sequencing, hidden rewards, reusable display systems, and custom packaging.
- Currently developing a *Sea of Thieves*-inspired sequential puzzle translating a digital game voyage into interconnected physical mechanisms.

Custom Product & Fixture Design

- Designed and manufactured functional products including a modular Dell desktop stand, adjustable trade-show structural connector, Arduino smart greenhouse, and controller display stand.
- Projects emphasize user requirements, manufacturability, compact packaging, serviceability, and real-world testing.

TECHNICAL SKILLS

CAD & Design: Fusion 360, Blender, mechanical assemblies, surface modeling, tolerance design, DFM

Fabrication: FDM 3D printing, ABS, laser cutting, waterjet-cut aluminum, hand fabrication, assembly

Electronics: KiCad, custom PCB layout, sensors, Arduino, wiring and connector integration

Programming: Java, Python, FTC SDK, OpenCV, TensorFlow, autonomous robotics

Engineering: Mechanism design, rapid prototyping, mechatronics, computer vision, troubleshooting, iterative testing

ACTIVITIES & RECOGNITION

FIRST Tech Challenge World Championship Competitor · NJ State Champion · Think Award

Personal “**You Inspire Me**” Judge Pin — FIRST World Championship

Varsity Swimming · Engineering outreach and STEM presentations